

# An Agent-Driven Pipeline for Mechanistic Interpretability of Single-Cell Foundation Models: Findings on MaxToki-217M and a Reusable Audit Framework

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## Abstract

We deploy an agent-driven mechanistic-interpretability pipeline against MaxToki-217M (Gómez Ortega et al., 2026), a recently released trajectory-aware single-cell foundation model, with eight pipelines executed end-to-end by Claude agents under ~15 hours of human supervision. The model encodes cell *identity geometry* but not directed *regulatory mechanism*: five methodologically distinct probes converge on a negative for fine-grained TF→target logic, while a hematopoietic developmental manifold is externally generalizable and compressible 2,400× to a 7.7 KB sparse operator, and activation steering at L0/L3/L9/L11 is causally effective with a non-monotonic mid-layer basin at L6. An audit-driven re-test inverts the original “0% TF specificity” headline at strict ChIP-seq threshold ( $p_{\text{null}}=0.030$ , post-hoc,  $n=1$ , awaits replication). We contribute (i) the agent-driven pipeline architecture, (ii) a reusable 10-pattern audit framework that detected three load-bearing flaws in our own runs, and (iii) the deployment study; all pipeline execution and audit were performed by AI agents.

## 1. Introduction

Single-cell foundation models (SCFM) such as scGPT (Cui et al., 2024), Geneformer (Theodoris et al., 2023), scFoundation (Hao et al., 2024), and the recently released MaxToki-217M (Gómez Ortega et al., 2026) are increasingly used for in-silico perturbation (Lotfollahi et al., 2019; Roohani et al., 2024), GRN inference, cell-type identification, and drug-target prioritization. Whether they actually *encode* the biology that downstream uses rely on, versus

merely re-organizing co-expression structure, is an open question (Kedzierska et al., 2023; Liu et al., 2023; Wenteler et al., 2024). Mechanistic interpretability addresses this directly: hypothesis-driven, intervention-supported analysis of what features inside a trained model represent and how they compose (Bricken et al., 2023; Conmy et al., 2023; Wang et al., 2023; Sharkey et al., 2025). A methodologically defensible mech-interp audit of one SCFM can take many person-months.

Recent autonomous-agent systems — the AI Scientist (Lu et al., 2024), Coscientist for autonomous chemistry (Boiko et al., 2023), ChemCrow (M. Bran et al., 2024), ML-engineering benchmarks like MLE-bench (Chan et al., 2024), and SWE-agent for software engineering (Yang et al., 2024) — show that LLM agents can carry out multi-step research with moderate-to-high autonomy. We investigate whether mech-interp on SCFMs can be largely automated. Eight mech-interp pipelines, originally developed in the authors’ prior work on scGPT and Geneformer (Anonymous, 2026a;f;g;e;b;d;c), are pre-authored as markdown executable specifications and executed end-to-end against MaxToki-217M by Claude Opus 4.7 agents under high-level human direction.

**Contributions.** (1) An *agent-executable pipeline architecture* in which markdown specifications act as executable contracts for LLM-based research agents. (2) A *recurring-issue audit framework* of ten reviewer-derived patterns codified as agent-applicable diagnostics; in this study the audit shifted three load-bearing verdicts. (3) A deployment study yielding six interrelated findings on MaxToki-217M that jointly characterize the model as encoding cell-identity geometry but not regulatory mechanism.

## 2. Related Work

**SCFM mech-interp.** Prior work on scGPT and Geneformer established the standard catalogue: per-layer attention extraction with TRRUST/STRING comparison (Anonymous, 2026a), residual-stream SVD geometry mapping subcellular localization and PPI networks (Anonymous,

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

2026f), topological hypothesis screening with hierarchical null models (Anonymous, 2026g), SAE atlases of the residual stream and causal circuit tracing (Anonymous, 2026e;b), and manifold discovery with extraction and compactification (Anonymous, 2026d). Convergent observation across this lineage: SCFMs reorganize co-expression structure and encode coarse cell-type identity but encode fine-grained regulatory logic poorly (Kedzierska et al., 2023; Liu et al., 2023; Wenteler et al., 2024).

**Mech-interp foundations.** Sparse autoencoders (Bricken et al., 2023; Cunningham et al., 2024; Templeton et al., 2024; Gao et al., 2024) and activation patching / circuit tracing (Conmy et al., 2023; Wang et al., 2023; Meng et al., 2022; Olsson et al., 2022) provide the technical machinery our pipelines deploy; sparse feature-circuit work (Marks et al., 2025; Lindsey et al., 2024) provides the closest comparator for the SAE-circuit-tracing pipeline. SAE-based mech-interp has been recently extended to protein language models (Simon & Zou, 2025), suggesting toolkit portability across biological domains. We adapt these tools at SCFM-specific scale:  $\sim 5,000$ -feature TopK SAEs per layer and  $\sim 2.1\text{M}$ -edge causal circuits.

**Autonomous research agents.** AutoGPT (Significant Gravitas, 2023), the AI Scientist (Lu et al., 2024), and domain-specific autonomous systems (Boiko et al., 2023; M. Bran et al., 2024) demonstrate multi-step LLM-driven research. Our agents build on ReAct/Reflexion/Toolformer (Yao et al., 2023; Shinn et al., 2023; Schick et al., 2023) but operate at longer scale (multi-day persistent-memory executions) against highly-specified human-authored pipelines. The agent-computer-interface idea in SWE-agent (Yang et al., 2024) is conceptually parallel: a structured contract shapes the moves the agent can make.

### 3. The Agent-Driven Pipeline Architecture

Figure 1 sketches the architecture. Three components are load-bearing: pipelines as markdown executable specifications, a per-target-model project directory layout, and a recurring-issue audit pipeline.

#### 3.1. Pipelines as executable specifications

Each method we deploy lives as a single markdown file in `pipelines/` following a frozen 10-section template: overview, source citation with pinned commit hash, inputs, outputs, dependencies, methodology with numbered steps, code references into pinned reference implementations, parameters table, validation signatures, and known pitfalls. The template is enforced; missing or out-of-order sections are rejected. The contract is strong enough that any suffi-

ciently capable LLM agent can execute the pipeline end-to-end without re-reading the source paper. Because reviewers’ recurring objections cluster around predictable omissions (missing null model, missing trivial baseline, missing positive control, missing scope qualifier), each gets a named slot with non-empty content; an agent that follows the template inherits much of the methodological hygiene by construction.

#### 3.2. Per-target-model project structure

For each SCFM, a directory under `projects/<target>/` holds tokenizer wiring, weight-download instructions, run artefacts, and per-pipeline summaries. Pipelines are model-agnostic; the project directory holds the model-specific glue. The split lets the same catalogue deploy against multiple SCFMs without modifying the pipelines, and lets new pipelines be added against an existing SCFM without re-extracting embeddings.

#### 3.3. Coordination and supervision

Most pipelines are single-agent. For pipelines with open-ended hypothesis search (the topology-141 pipeline, the manifold-discovery sweep), an autonomous loop alternates “brainstormer” and “executor” turns; we adopt a **2-consecutive-negatives retirement rule** to prevent the loop from re-testing variants of an exhausted hypothesis. Persistent file-based memory across conversations is essential for multi-day runs. The supervisor issues run requests, specifies chance-baseline expectations before launch, and gates expensive operations, but does not write extraction code, debug runs, or compose summaries. Across the full MaxToki-217M project, human supervision totalled  $\sim 15$  hours over three weeks against  $\sim 50$  GPU-hours of agent compute.

#### 3.4. The recurring-issue audit pipeline

We extracted ten recurring methodological objections from six independent peer-reviewer comment sets in the authors’ prior work and codified them as a reusable checklist (Table 1). Each pattern has a diagnostic signature, a prescription, and a citation back to the review documents that established the pattern. The audit runs end-to-end on a project folder in 1–2 days of agent time and produces a per-pattern  $8 \times 10$  verdict matrix and an action-proposal table. The audit itself is executed autonomously by an agent against artefacts produced by prior agent runs, so its validity rests on the checklist being more reliable than any individual run — an assumption we revisit in §7. The audit has been validated only on this project; transfer to held-out research projects remains open.

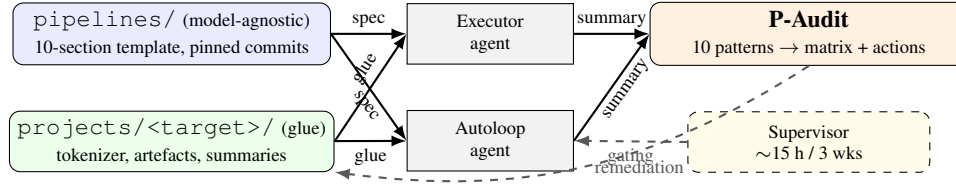


Figure 1. The agent-driven pipeline architecture. Markdown specifications under `pipelines/` act as executable contracts for Claude agents; per-target-model glue lives under `projects/<target>/`; the audit pipeline runs after each batch, applying ten reviewer-derived patterns to produced artefacts and emitting an action-proposal table. The supervisor issues run requests, sets chance-baseline expectations, and gates expensive operations.

Table 1. The 10-pattern audit checklist. P1–P10 codify recurring peer-reviewer objections from six independent reviews of papers in the authors’ prior work.

| Pattern | Description                           |
|---------|---------------------------------------|
| P1      | Weak null / not anchored to chance    |
| P2      | Trivial / gene-level baseline absent  |
| P3      | Endpoint–object mismatch              |
| P4      | Pseudoreplication / leaky CV          |
| P5      | Selection-driven inflation            |
| P6      | No positive control for null findings |
| P7      | Single-condition generalization       |
| P8      | Stability of headline numbers         |
| P9      | Causal language overreach             |
| P10     | Reproducibility scaffolding           |

Table 2. Mech-interp pipelines deployed against MaxToki-217M. Wall clock is agent-side end-to-end; “+ ext.” denotes hypothesis-extension phases beyond the base pipeline.

| Pipeline            | Probes              | Hr.  |
|---------------------|---------------------|------|
| spectral-geometry   | residual SVD geom.  | 0.5  |
| attention-grn (x4)  | attention→GRN       | 5.5  |
| topology-141        | 141-hyp. topology   | 4+   |
| sae-atlas           | TopK SAEs ~5K/L     | 2.5  |
| sae-circuit-tracing | 2.1M-edge circuit   | 4    |
| sae-exhaustive-map. | 4.97M edges + steer | 5.5+ |
| manifold-discovery  | H65 hematopoietic   | 24   |
| longevity-mechint.  | donor-aware aging   | 1.5  |

methodologically clean while exposing a cross-architectural-family scope qualifier (Section 5.5).

## 4. Application: MaxToki-217M

### 4.1. Why MaxToki-217M

MaxToki-217M was selected as the application target for three reasons. (1) *Frontier model*. It is the most recent major SCFM release at the time of writing (March 2026), from the Theodoris/NVIDIA team, scaling pre-training to ~1T gene tokens across ~175M human single-cell transcriptomes spanning birth through 90+ years (Gómez Ortega et al., 2026). (2) *New data and training regime*. Unlike scGPT (encoder-only, rank-binned values) or Geneformer (encoder-only, rank-ordered Ensembl IDs), MaxToki-217M is an autoregressive Llama-architecture transformer trained in two stages, generation followed by trajectory modelling. Context length is therefore a *trajectory-length* budget — a cell occupies one position — not a per-cell gene-count budget; this breaks the inherited Geneformer convention of  $\leq 2,048$  genes per cell, and tests whether pipelines built on encoder-only SCFMs port to the autoregressive trajectory regime. (3) *An external port for our catalogue*. The eight pipelines were developed against scGPT and Geneformer; MaxToki-217M is the first target outside that family and serves as a stress-test for the “model-agnostic specification” claim. Crucially, MaxToki-217M shares its 20,275-token Ensembl-ID gene tokenizer exactly with Geneformer V2-316M, which makes within-family cross-model alignment

### 4.2. Target architecture

MaxToki-217M is a Llama-architecture autoregressive transformer (11 layers, hidden size 1232, 8 attention heads/layer, vocabulary 20,275, context 4,096 tokens). The original paper’s analyses focus on cell-state trajectory prediction and steering; ours target the model’s internal representations directly via attention, residual-stream, and SAE-feature probes.

### 4.3. Pipelines deployed

We deploy eight pipelines against MaxToki-217M (Table 2). The project-specific glue (gene-symbol-to-token mapping, hidden-state extraction harness, dataset paths) lives under `projects/maxtoki/setup/`.

## 5. Findings

A unifying picture emerges: MaxToki-217M encodes cell *identity geometry* that is recoverable, externally generalizable, and locally compressible, but encodes fine-grained *regulatory mechanism* only weakly. The convergent regulatory logic negative (§5.1), the manifold positive (§5.2), and the GATA1 audit-driven re-test (§5.3) triangulate this distinction.

### 5.1. Convergent negative on regulatory logic

Five methodologically distinct probes agree that MaxToki-217M does not encode fine-grained TF→target regulatory logic at a resolution useful for in-silico perturbation. The convergence is itself the only positive control available for a negative finding: no single methodological flaw plausibly pushes all five in the same direction. We note shared assumptions across the probes (model forward passes, Ensembl-ID tokenizer, partly overlapping ground-truth databases) and return to common-mode failure modes in §7.

- **attention→GRN**: Curveball degree-preserving null  $z \approx 0$  across four runs (K562, RPE1, Adamson, MaxToki-1B) (Replogle et al., 2022); a 1-feature gene-variance baseline beats attention by 6–20 AUROC points ( $p_{BH} \leq 10^{-4}$ ); residualized AUROC collapses to chance.
- **circuit-tracing** CRISPRi validation: directional accuracy 53.48% overall ( $n=1,503,408$ ); per-source mean 53.42% (95% CI [52.60%, 54.16%]) under GroupKFold-by-source-gene 5-fold. Real but small.
- **sae-atlas** causal-patching specificity: median  $1.01\times$  (paper Geneformer:  $2.36\times$ ); 0/48 TFs “specific” under TRRUST — but see §5.3.
- **topology-141** signed motif-community (H123): 0–1/12 layers with positive null gap per domain; under strict dual-axis disjoint CV, immune AUROC drops  $0.505 \rightarrow 0.462$ .
- **exhaustive-mapping** combinatorial ablation: zero higher-order synergy (0/2980 superadditive triplets); three-way redundancy ratio 0.190 (paper: 0.59).

### 5.2. Manifold discovery as a deployable positive

MaxToki-217M’s residual stream is a multi-axis biological hub. A 7.7 KB hard-sparse operator (16 features  $\times$  60 genes) extracts a hematopoietic developmental ordering (H65) that passes four pre-registered quality gates: trustworthiness 0.811 (paired within-branch shuffle null 0.799); branch-holdout Spearman 0.370 (null  $-0.001$ ); donor-holdout 0.013; permutation  $-0.016$ . The branch-holdout gap (0.370 vs.  $-0.001$ ) is the discriminating signal. On 13 disjoint donors, trustworthiness is 0.896 (95% CI [0.884, 0.899]); zero-shot transfer to 12 multi-tissue donors yields 0.827. A non-hematopoietic lung negative control fails 3 of 4 gates decisively. Compactification reduces a 154-dim full-drift operator to the 7.7 KB sparse form (2,400 $\times$  compression) at no quality loss. A sweep over 12 candidate biological rulers finds two further externally-transferring manifolds (H95 effector modality, H103 B-cell maturation) plus one internally-passing (H38 signalling, external transfer not tested).

The mechanistic substrate is *distributed*: top-4 SVD factors of the L10H6 head explain only 18.3% of pooled ablation

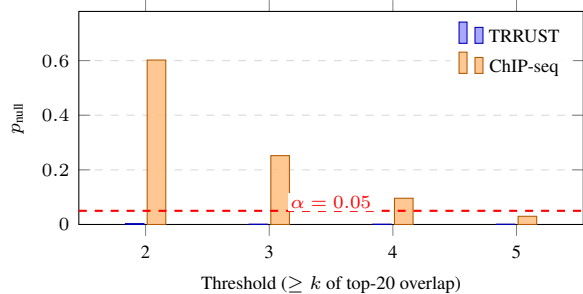


Figure 2. GATA1 feature-specificity  $p_{null}$  under TRRUST (narrow ground truth, 57 targets) vs. dorothea ChIP-seq direct targets (470 targets) across thresholds. TRRUST is near-zero by construction (the headline “0/48” was underpowered); ChIP-seq at strict  $\geq 5$  threshold yields  $p_{null} = 0.030$  for  $n=1$  TF. The threshold was selected post-hoc; under Bonferroni across the four thresholds ( $\alpha/4=0.0125$ ) the result is no longer significant. Replication on additional cell lines and TFs is required.

impact (paper scGPT: 66.2%); a top-4 core-sufficiency test collapses branch from 0.973 to 0.123. The geometry is real and externally generalizable but does not concentrate in a compact mechanistic core.

### 5.3. Audit-driven re-test of TF specificity

The sae-atlas TF-specificity test originally reported 0/48 specific under TRRUST: a TF is specific if at least one significantly-responding SAE feature has  $\geq 2$  of its top-20 genes in the TF’s TRRUST target set. Audit pattern P6 (no positive control) flagged this as underpowered: GATA1 has only 57 TRRUST targets and TAL1 has 10, making the threshold near-impossible under random feature draws. Re-running the test with feature IDs retained and evaluating against dorothea ChIP-seq direct targets (García-Alonso et al., 2019) at thresholds {2, 3, 4, 5}-of-top-20 (Figure 2), GATA1’s responding feature 2610 has 8 of its top-20 genes overlapping the 470-element ChIP-seq direct-target set, with  $p_{null} = 0.030$  over 1,000 random-feature draws (Phipson & Smyth, 2010).

**Caveats.** The result rests on a single TF (GATA1), in a single cell line (Replogle K562 (Replogle et al., 2022)), against a single ChIP-seq dataset, with the threshold selected post-hoc. MYC and TAL1 are not perturbed in K562 and could not be tested. The right reading is narrow: the original 0% verdict is substantially an artefact of TRRUST narrowness, and under proper ChIP-seq ground truth there is at most suggestive evidence ( $p = 0.030$  uncorrected) that MaxToki-217M’s SAE features encode TF-target regulatory information — subject to replication.

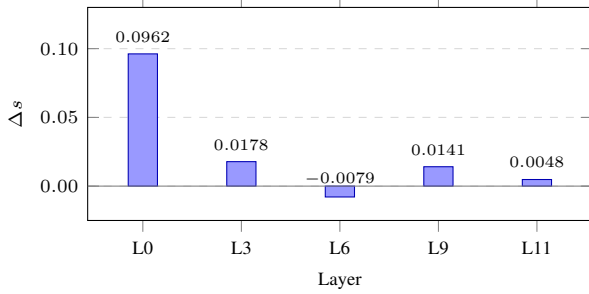


Figure 3. Activation-steering effect  $\Delta s$  across MaxToki-217M layers. L0 has the largest push toward maturity ( $\sim 20\times$  larger than L11); L6 inverts to counter-differentiation. L9 and L11 are positive but small; we report them as suggestive (no permutation null over feature selection was bounded). Unlike Geneformer’s monotone L0→L17 progression (Anonymous, 2026c), MaxToki-217M shows a non-monotonic mid-layer basin — consistent with trajectory-aware pre-training distributing pseudotemporal information throughout the stack.

#### 5.4. Trajectory steering and cell-state control

The exhaustive-mapping pipeline’s trajectory-steering experiment provides the project’s first *causal positive*: activation addition along selected SAE features in early-pseudotime Tabula Sapiens immune cells (The Tabula Sapiens Consortium, 2022) pushes cell-state logits toward maturity. Using the spec-compliant  $\Delta s = \cos(z', g_{\text{late}}) - \cos(z', g_{\text{early}}) - \text{baseline metric}$ ,  $\Delta s$  is positive at L0/L3/L9/L11 and inverts to a counter-differentiation basin at L6 (Figure 3). Top-upregulated genes are biologically coherent: L0 upregulates SOD3 and KITLG (progenitor / stemness markers), L6 upregulates C1QB (myeloid complement), L11 upregulates FTL (terminal erythroid ferritin).

The L0 effect ( $\sim 20\times$  larger than L11) is the clearest positive; the L9 and L11 magnitudes are near a noise floor for which we have not bounded a separate permutation null over feature selection, and are reported as suggestive. The finding is consistent with the regulatory-logic negative (§5.1): MaxToki-217M represents cell *identity* along a differentiation axis, not the *regulatory mechanism* that moves a cell along it.

#### 5.5. Cross-architectural scope and cross-layer stability

Cross-model alignment with Geneformer V2-316M replicates strongly: CCA  $r=0.78$ , pairwise gene-similarity Pearson 0.382 (95% CI [0.380, 0.384]). Alignment with scGPT breaks: CCA  $r=0.40$ , pairwise Pearson  $\approx 0$  (slightly negative on immune,  $z=-3.0$ ), top-1 retrieval 5–8% (vs. 38–47% against Geneformer V2-316M). scGPT differs from MaxToki-217M in both tokenization and architecture, so the failure isolates to architectural-family scope.

Within MaxToki-217M, gene-pair geometry is moderately stable across layers, depending on the feature construction. Cross-layer Pearson is 0.604–0.678 for the H123 signed motif-community feature and 0.561 for Euclidean distance, but only 0.275 for the triangle-defect feature. A correlation of  $\sim 0.65$  shares  $\sim 40\%$  of variance, so the layers do not encode identical gene-pair geometry — the framing is “moderate conservation,” not “no change.” Within-layer stability and within-family cross-model agreement appear to project the same observation: SCFM input embeddings carry most of the gene-pair geometric load.

## 6. Audit-Detected Failures

Two remediation cycles raised the share of fully addressed audit patterns from 61% to 75% across the eight pipelines. Three audit-detected failures shifted headline numbers materially: a sign-tracking error in the original directional-accuracy implementation (mapping to P3, endpoint–object mismatch); a TF-specificity test reported as 0/48 without anchoring to a chance baseline that was near-zero by construction (P1/P6); and several headline numbers reported as point estimates without bootstrap CIs (P8). Without the audit pass, this paper would have reported “0/48 TF specificity” as a clean negative and “54.61% directional accuracy” as a positive — one verdict-level overconfident in each direction. We have not formally validated that headlines the audit did *not* flag are equally reliable, a known limit of an agent-policed-by-agent setup (§7).

## 7. Discussion

**What worked.** The pipeline-as-spec contract held; agents executed pipelines without re-reading source papers. The template caught structural omissions. Persistent file-based memory was essential for multi-day runs. The 2-consecutive-negatives retirement rule prevented exhausted-hypothesis loops. The audit detected three load-bearing flaws.

**Common-mode risks for the convergent negative.** The five probes that converge on the regulatory-logic negative are not strictly independent: they share model forward passes, the Ensembl-ID tokenizer, partial overlap in ground-truth databases, and were developed by the same authors. A single shared confound — e.g., training data weighted away from perturbation experiments, or systematic gene-coverage gaps in the ground-truth resources — could push several probes the same way. The convergence is strong evidence within the assumptions the probes share; it is not evidence *about* those assumptions.

**Generalization.** The catalogue ports cleanly between SCFMs sharing the standard `transformers` interface;

MaxToki-217M is the first port outside the original scGPT / Geneformer family. The audit framework is architecture-independent at the pattern level. Non-SCFM biological foundation models (Lin et al., 2023; Jumper et al., 2021; Nguyen et al., 2024; Dalla-Torre et al., 2024) remain untested; recent SAE work on protein language models (Simon & Zou, 2025) suggests a plausible path.

**Limitations.** The agent system runs the human-authored catalogue but does not generate novel pipeline architectures. The audit framework has been validated on a single project. Three audit categories were not closed: a cell-level robustness comparison against scVI / Palantir / CellTypist baselines; longevity probing at AIDA scale; and cross-architectural replication of the H123 probe on scGPT. The GATA1 ChIP-seq inversion rests on  $n=1$  TF in K562 at a post-hoc-selected threshold ( $p=0.030$  uncorrected, marginal under Bonferroni); replication is the next required step. We have not run a non-agent baseline on the same pipelines, so the methodological contribution is documented as a deployment study with an audit case study, not a head-to-head against human or alternative-agent execution.

## 8. Conclusion

We applied an agent-driven mech-interp architecture to MaxToki-217M, a recently released trajectory-aware single-cell foundation model. Approximately 15 hours of high-level human supervision direct  $\sim 50$  GPU-hours of agent compute through a methodologically defensible mech-interp pass yielding six interrelated findings that jointly characterize MaxToki-217M as encoding cell-identity geometry but not directed regulatory mechanism. The pipeline catalogue itself originates in prior work; the contribution here is the agent-execution architecture, the reusable audit framework, and the deployment study.

**Reproducibility.** Run artefacts, audit reports, and per-pipeline summaries are released at the project repository (URL withheld for anonymous review). MaxToki-217M, scGPT, and Geneformer V2-316M weights are taken from public HuggingFace releases at revisions pinned in `projects/<target>/setup/`. Random seeds, dataset versions, software versions, and per-pipeline hardware (a single A100-80GB for forward passes; CPU for SAE training and analysis) are recorded in each run’s `run.log`.

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